

Numerical Methods

By

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Chapter (4): Numerical Differentiation and Integration

Numerical Differentiation

Numerical differentiation is a method to compute the derivatives of a function at some values of independent variable x , when the function $f(x)$ is explicitly unknown, however it is known only for a set of arguments.

we approximate the derivative using finite differences.

Derivatives formulas:

Numerical Differentiation for Equally Spaced Data

- Derivatives based on **Newton's forward** interpolation formula. This formula is used to find the derivative for some given x lying **near the beginning of the data table**.
- Derivatives based on **Newton's backward** interpolation formula. This formula is suitable to find the derivative for a point **near the end of the data table**.
- Derivatives based on **Stirling's interpolation** formula. This formula is used to find the derivative for some point lying **near the middle of the tabulated value**.

Derivatives formulas:

Numerical Differentiation for Unequally Spaced Data

- Lagrange's interpolation formula.

- ❑ In numerical differentiation, sometimes we do not know the explicit formula of a function $f(x)$, but we have its values in a table.
- ❑ If the x -values are equally spaced, we can use Newton's Forward Interpolation Formula and then differentiate it to obtain formulas for derivatives.
- ❑ This method is useful when the value of the derivative is required near the beginning of the table

General Formula for the First Derivative:

$$f'(x) = \frac{1}{h} \left[\Delta y_0 + \frac{2p-1}{2!} \Delta^2 y_0 + \frac{3p^2-6p+2}{3!} \Delta^3 y_0 + \frac{4p^3-18p^2+22p-6}{4!} \Delta^4 y_0 + \frac{5p^4-40p^3+105p^2-100p+24}{5!} \Delta^5 y_0 - \dots \right]$$

Where $p = \frac{x-x_0}{h}$

First Derivative Formula at the point $x = x_0$:

When $x = x_0$, $p = 0$, Therefore

$$f'(x_0) = \frac{1}{h} \left[\Delta y_0 - \frac{1}{2} \Delta^2 y_0 + \frac{1}{3} \Delta^3 y_0 - \frac{1}{4} \Delta^4 y_0 + \frac{1}{5} \Delta^5 y_0 - \dots \right]$$

General Formula for the Second Derivative:

$$f''(x) = \frac{1}{h^2} \left[\Delta^2 y_0 + \frac{6p-6}{3!} \Delta^3 y_0 + \frac{12p^2-36p+22}{4!} \Delta^4 y_0 + \frac{20p^3-120p^2+210p-100}{5!} \Delta^5 y_0 - \dots \right]$$

Where $p = \frac{x-x_0}{h}$

Second Derivative Formula at the point $x = x_0$:

When $x = x_0$, $p = 0$, Therefore

$$f''(x_0) = \frac{1}{h^2} \left[\Delta^2 y_0 - \Delta^3 y_0 + \frac{11}{12} \Delta^4 y_0 - \frac{5}{6} \Delta^5 y_0 + \dots \right]$$

Example :

From the following table find the value of $f'(x)$ and $f''(x)$ at the point $x = 1.0$

x	1	1.1	1.2	1.3	1.4	1.5
y	5.4680	5.6665	5.9264	6.2551	6.6601	7.1488

Solution :

- $x = 1.0$ is a tabulated point near the beginning of the table
- Forward difference table

x	y	Δy	$\Delta^2 y$	$\Delta^3 y$	$\Delta^4 y$	$\Delta^5 y$
1	5.4680					
		0.1985				
1.1	5.6665		0.0614			
		0.2599		0.0074		
1.2	5.9264		0.0688		0	
		0.3287		0.0074		0
1.3	6.2551		0.0763		0	
		0.4050		0.0074		
1.4	6.6601		0.0837			
		0.4887				
1.5	7.1488					

$$f'(x_0) = \frac{1}{h} \left[\Delta y_0 - \frac{1}{2} \Delta^2 y_0 + \frac{1}{3} \Delta^3 y_0 - \frac{1}{4} \Delta^4 y_0 + \frac{1}{5} \Delta^5 y_0 - \dots \right]$$

$$h = 0.1$$

$$f'(1) = \frac{1}{0.1} \left[0.1985 - \frac{1}{2} (0.0614) + \frac{1}{3} (0.0074) - \frac{1}{4} (0) + \frac{1}{5} (0) \right] = 1.7020$$

$$f''(x_0) = \frac{1}{h^2} \left[\Delta^2 y_0 - \Delta^3 y_0 + \frac{11}{12} \Delta^4 y_0 - \frac{5}{6} \Delta^5 y_0 + \dots \right]$$

$$f''(1) = \frac{1}{(0.1)^2} \left[0.0614 - 0.0074 + \frac{11}{12} (0) - \frac{5}{6} (0) \right] = 5.4040$$

Example :

Obtain the first and second derivatives of the function tabulated below at the points $x = 1.2$.

x	1	1.2	1.4	1.6	1.8	2
y	0	0.128	0.544	1.298	2.44	4.02

Solution :

- $x = 1.2$ is a tabulated point near the beginning of the table
- Forward difference table

x	y	Δy	$\Delta^2 y$	$\Delta^3 y$	$\Delta^4 y$	$\Delta^5 y$
1	0					
		0.128				
1.2	0.128		0.288			
		0.416		0.05		
1.4	0.544		0.338		0	
		0.754		0.05		0
1.6	1.298		0.388		0	
		1.142		0.05		
1.8	2.440		0.438			
		1.580				
2	4.02					

$$f'(x_0) = \frac{1}{h} \left[\Delta y_0 - \frac{1}{2} \Delta^2 y_0 + \frac{1}{3} \Delta^3 y_0 - \frac{1}{4} \Delta^4 y_0 + \frac{1}{5} \Delta^5 y_0 - \dots \right]$$

$$h = 0.1$$

$$f'(1.2) = \frac{1}{0.2} \left[0.416 - \frac{1}{2} (0.338) + \frac{1}{3} (0.05) - \frac{1}{4} (0) + \frac{1}{5} (0) \right] = 1.31833$$

$$f''(x_0) = \frac{1}{h^2} \left[\Delta^2 y_0 - \Delta^3 y_0 + \frac{11}{12} \Delta^4 y_0 - \frac{5}{6} \Delta^5 y_0 + \dots \right]$$

$$f''(1) = \frac{1}{(0.2)^2} \left[0.338 - 0.05 + \frac{11}{12} (0) - \frac{5}{6} (0) \right] = 7.2$$

Example :

Obtain the first and second derivatives of the function tabulated below at the points $x = 1.2$.

x	1	1.2	1.4	1.6	1.8	2
y	0	0.1	0.5	1.25	2.4	3.9

Solution :

Try by yourself

Derivatives based on Newton's backward interpolation

- Newton's Backward Interpolation Formula is used when the required value is near the end of the table and the x -values are equally spaced.

General Formula for the First Derivative:

$$f'(x) = \frac{1}{h} \left[\nabla y_n + \frac{2p+1}{2!} \nabla^2 y_n + \frac{3p^2+6p+2}{3!} \nabla^3 y_n + \frac{4p^3+18p^2+22p+6}{4!} \nabla^4 y_n + \frac{5p^4+40p^3+105p^2+100p+24}{5!} \nabla^5 y_n + \dots \right]$$

Where $p = \frac{x-x_n}{h}$

First Derivative Formula at the point $x = x_n$:

When $x = x_n$, $p = 0$, Therefore

$$f'(x_n) = \frac{1}{h} \left[\nabla y_n + \frac{1}{2} \nabla^2 y_n + \frac{1}{3} \nabla^3 y_n + \frac{1}{4} \nabla^4 y_0 + \frac{1}{5} \nabla^5 y_n + \dots \right]$$

General Formula for the Second Derivative:

$$f''(x) = \frac{1}{h^2} \left[\nabla^2 y_n + \frac{6p+6}{3!} \nabla^3 y_n + \frac{12p^2+36p+22}{4!} \nabla^4 y_n + \frac{20p^3+120p^2+210p+100}{5!} \nabla^5 y_n + \dots \right]$$

Where $p = \frac{x-x_n}{h}$

Second Derivative Formula at the point $x = x_n$:

When $x = x_n$, $p = 0$, Therefore

$$f''(x_n) = \frac{1}{h^2} \left[\nabla^2 y_n + \nabla^3 y_n + \frac{11}{12} \nabla^4 y_n + \frac{5}{6} \nabla^5 y_n + \dots \right]$$

Example :

Obtain the first and second derivatives of the function tabulated below at the points $x = 6$.

x	1	2	3	4	5	6
y	0.0201	0.0844	0.3444	1.0100	2.3660	4.7719

Solution :

- $x = 6$ is a tabulated point near the end of the table
- backward difference table

x	y	∇y	$\nabla^2 y$	$\nabla^3 y$	$\nabla^4 y$	$\nabla^5 y$
1	0.0201					
		0.0643				
2	0.0844		0.1957			
		0.2600		0.2100		
3	0.3444		0.4056		0.0748	
		0.6656		0.2847		0
4	1.0100		0.6904		0.0748	
		1.3560		0.3595		
5	2.3660		1.0499			
		2.4059				
6	4.7719					

$$f'(x_n) = \frac{1}{h} \left[\nabla y_n + \frac{1}{2} \nabla^2 y_n + \frac{1}{3} \nabla^3 y_n + \frac{1}{4} \nabla^4 y_0 + \frac{1}{5} \nabla^5 y_n + \dots \right]$$

$$h = 1$$

$$f'(6) = \frac{1}{1} \left[2.4059 + \frac{1}{2} (1.0499) + \frac{1}{3} (0.3595) + \frac{1}{4} (0.0748) + \frac{1}{5} (0) \right] = 3.0694$$

$$f''(x_n) = \frac{1}{h^2} \left[\nabla^2 y_n + \nabla^3 y_n + \frac{11}{12} \nabla^4 y_n + \frac{5}{6} \nabla^5 y_n + \dots \right]$$

$$f''(6) = \frac{1}{(1)^2} \left[1.0499 + 0.3595 + \frac{11}{12} (0.0748) + \frac{5}{6} (0) \right] = 1.4780$$

Example :

Obtain the first and second derivatives of the function tabulated below at the points $x = 2$.

x	1	1.2	1.4	1.6	1.8	2
y	0	0.1	0.5	1.25	2.4	3.9

Solution :

Try by yourself

Derivatives based on Stirling's interpolation

- ❑ Stirling's interpolation Formula is used when the required value is near the center of the table and the x -values are equally spaced.

General Formula for the First Derivative:

$$f'(x) = \frac{1}{h} \left[\frac{\Delta y_{-1} + \Delta y_0}{2} + p \Delta^2 y_{-1} + \frac{3p^2 - 1}{6} \left(\frac{\Delta^3 y_{-2} + \Delta^3 y_{-1}}{2} \right) + \frac{2p^3 - p}{12} \Delta^4 y_{-2} \right. \\ \left. + \frac{5p^4 - 15p^2 + 4}{120} \left(\frac{\Delta^5 y_{-3} + \Delta^5 y_{-2}}{2} \right) + \dots \right]$$

Where $p = \frac{x - x_0}{h}$

First Derivative Formula at the point $x = x_0$:

When $x = x_0$, $p = 0$, Therefore

$$f'(x_0) = \frac{1}{h} \left[\frac{\Delta y_{-1} + \Delta y_0}{2} - \frac{1}{6} \left(\frac{\Delta^3 y_{-2} + \Delta^3 y_{-1}}{2} \right) + \frac{1}{30} \left(\frac{\Delta^5 y_{-3} + \Delta^5 y_{-2}}{2} \right) + \dots \right]$$

General Formula for the Second Derivative:

$$f''(x) = \frac{1}{h^2} \left[\Delta^2 y_{-1} + p \left(\frac{\Delta^3 y_{-2} + \Delta^3 y_{-1}}{2} \right) + \frac{6p^2 - 1}{12} \Delta^4 y_{-2} \right. \\ \left. + \frac{2p^3 - 3p}{12} \left(\frac{\Delta^5 y_{-3} + \Delta^5 y_{-2}}{2} \right) + \dots \right]$$

Where $p = \frac{x - x_0}{h}$

First Derivative Formula at the point $x = x_0$:

When $x = x_0$, $p = 0$, Therefore

$$f''(x_0) = \frac{1}{h^2} \left[\Delta^2 y_{-1} - \frac{1}{12} \Delta^4 y_{-2} + \dots \right]$$

Example :

Obtain the first and second derivatives of the function tabulated below at the points $x = 0.2$.

x	0	0.1	0.2	0.3	0.4	0.5
y	0	0.10017	0.20134	0.30452	0.41076	0.52115

Solution :

- $x = 0.2$ is a tabulated point near the center of the table
- Central difference table

x	y	Δy	$\Delta^2 y$	$\Delta^3 y$	$\Delta^4 y$	$\Delta^5 y$
$x_{-2} = 0$	0					
		0.10017				
$x_{-1} = 0.1$	0.10017		0.001			
		0.10117		0.00101		
$x_0 = 0.2$	0.20134		0.00201		0.00004	
		0.10318		0.00105		0
$x_1 = 0.3$	0.30452		0.00306		0.00004	
		0.10624		0.00109		
$x_2 = 0.4$	0.41076		0.00415			
		0.11039				
$x_3 = 0.5$	0.52115					

Central difference table

$$f'(x_0) = \frac{1}{h} \left[\frac{\Delta y_{-1} + \Delta y_0}{2} - \frac{1}{6} \left(\frac{\Delta^3 y_{-2} + \Delta^3 y_{-1}}{2} \right) + \frac{1}{30} \left(\frac{\Delta^5 y_{-3} + \Delta^5 y_{-2}}{2} \right) + \dots \right]$$

$h = 0.1$

$$f'(0.2) = \frac{1}{0.1} \left[\frac{0.10117 + 0.10318}{2} - \frac{1}{6} \left(\frac{0.00101 + 0.00105}{2} \right) \right] = 1.020033$$

$$f''(x_0) = \frac{1}{h^2} \left[\Delta^2 y_{-1} - \frac{1}{12} \Delta^4 y_{-2} + \dots \right]$$

$$f''(0.2) = \frac{1}{(0.1)^2} \left[0.00201 - \frac{1}{12} (0.00004) \right] = 0.200666$$

Example :

Obtain the first and second derivatives of the function tabulated below at the points $x = 3$.

x	1	2	3	4	5
y	0	1.4	3.3	5.6	8.1

Solution :

Try by yourself

Derivatives based on Lagrange's interpolation

- ❑ Lagrange's interpolation Formula is used when the x -values are unequally spaced.

Derivatives formulas:

Lagrange's interpolation formula.

$$\begin{aligned} f(x) &= \frac{(x - x_1)(x - x_2) \dots (x - x_n)}{(x_0 - x_1)(x_0 - x_2) \dots (x_0 - x_n)} y_0 \\ &+ \frac{(x - x_0)(x - x_2) \dots (x - x_n)}{(x_1 - x_0)(x_1 - x_2) \dots (x_1 - x_n)} y_1 + \dots \\ &+ \frac{(x - x_0)(x - x_1) \dots (x - x_{n-1})}{(x_n - x_0)(x_n - x_1) \dots (x_n - x_{n-1})} y_n \end{aligned}$$

Differentiating both sides w.r.t. x , we get $f'(x)$

Example :

Obtain the first and second derivatives of the function tabulated below at the points $x = 6$.

x	5	6	9	11
y	380	-2	196	508

Solution :

The Lagrange's interpolation formula is:

$$\begin{aligned}
 &f(x) \\
 &= \frac{(x - x_1)(x - x_2)(x - x_3)}{(x_0 - x_1)(x_0 - x_2)(x_0 - x_3)} y_0 + \frac{(x - x_0)(x - x_2)(x - x_3)}{(x_1 - x_0)(x_1 - x_2)(x_1 - x_3)} y_1 \\
 &+ \frac{(x - x_0)(x - x_1)(x - x_3)}{(x_2 - x_0)(x_2 - x_1)(x_2 - x_3)} y_2 + \frac{(x - x_0)(x - x_1)(x - x_2)}{(x_3 - x_0)(x_3 - x_1)(x_3 - x_2)} y_3
 \end{aligned}$$

$$x_0 = 5, x_1 = 6, x_2 = 9, x_3 = 11$$

$$y_0 = 380, y_1 = -2, y_2 = 196, y_3 = 508$$

$$f(x)$$

$$= \frac{(x-6)(x-9)(x-11)}{(5-6)(5-9)(5-11)}(380) + \frac{(x-5)(x-9)(x-11)}{(6-5)(6-9)(6-11)}(-2)$$

$$+ \frac{(x-5)(x-6)(x-11)}{(9-5)(9-6)(9-11)}(196) + \frac{(x-5)(x-6)(x-9)}{(11-5)(11-6)(11-9)}(508)$$

$$f(x) = \frac{-47}{3}x^3 + \frac{1276}{3}x^2 - 3635x + 9880$$

$$f'(x) = -47x^2 + \frac{2552}{3}x - 3635$$

$$f''(x) = -94x + \frac{2552}{3}$$

$$f'(6) = -47x^2 + \frac{2552}{3}x - 3635 = -223$$

$$f''(6) = -94x + \frac{2552}{3} = \frac{860}{3}$$

Example :

Obtain the first and second derivatives of the function tabulated below at the points $x = 3$.

x	2	3	5
y	4	9	25

Solution :

Try by yourself

**THANK YOU
FOR YOUR
ATTENTION**

